Name ____ Date ___________ Period _____

Continuous vs. Discrete

TEACHER KEY | Algebra II | Unit 1 | Day 3 | TEKS A2.2.A, A2.7.I

Objective: I can decide whether a domain and range are continuous or discrete, and write each one using the notation that fits.

Vocabulary

Continuous — every value between two points is included. The graph is an **unbroken** curve or line.

Discrete — only certain separated values count. The graph is **isolated points**.

Whole numbers — **0, 1, 2, 3, ...** **Integers** — **..., -2, -1, 0, 1, 2, ...**

Part 1 — Sort the Situations

Write C for continuous or D for discrete.

Situation	C or D	Why?
The number of chairs in a classroom	**D**	**no half chairs**
The distance a car travels	**C**	**any distance is possible**
Your age measured in years and days	**C**	**time flows without gaps**
The number of pets a family owns	**D**	**whole animals only**
The weight of a watermelon	**C**	**any weight is possible**

Part 2 — The Same Function, Two Ways

A cube-shaped box has side x inches, so its volume is **$V(x) = x^3$**.

x	1	2	3	4
$V(x) = x^3$	1	8	27	64

Case A: the box can be any size at all.
Domain **$x > 0$, $(0, \infty)$** Range **$V > 0$, $(0, \infty)$** **continuous**

Case B: the box is built out of 1-inch cube blocks, so x must be a whole number.
Domain **$x = 1, 2, 3, \ldots$** Range **$\{1, 8, 27, 64, \ldots\}$** **discrete**

Same equation, different **situation** → different domain. The context decides.

Part 3 — Which Notation Fits?

Continuous domains use **interval** notation, like [0, 5].

Discrete domains get **listed** in braces, like { 0, 1, 2, 3 }, because there are gaps between the values.

You Try 1. A stadium sells 0 to 5 hot dogs per person. Domain **{ 0, 1, 2, 3, 4, 5 }**

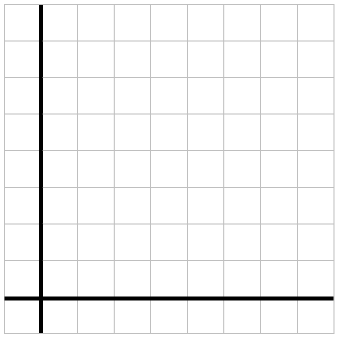
You Try 2. A candle burns from 10 cm down to 0 cm. Domain of heights **[0, 10]**

You Try 3. Why can You Try 1 not use interval notation? **it is discrete - values have gaps**

Part 4 — Graph Both Kinds

Each square is 1 unit. Plot each, then name the domain and range.

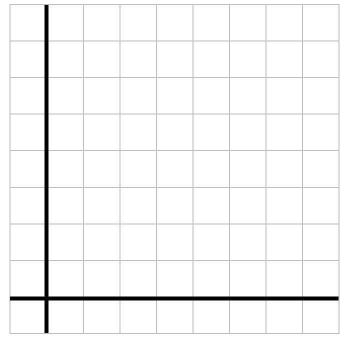
A. Discrete. A tutor charges $2 per session, up to 4 sessions. Plot (1, 2), (2, 4), (3, 6), (4, 8).



Domain **{ 1, 2, 3, 4 }**

Range **{ 2, 4, 6, 8 }**

B. Continuous. A candle burns 2 cm per hour from 8 cm. Graph $h = 8 - 2t$ for $0 \le t \le 4$.



Domain **[0, 4]**

Range **[0, 8]**

Part 5 — Quick Check

1. Should a discrete domain ever be written as an interval? **no**
2. Can a continuous range include negative numbers? **yes**
3. Is money spent at a vending machine continuous? **no - it jumps by coin amounts**

Practice

Decide continuous or discrete, then write the domain in the notation that fits.

1. A runner covers any distance from 0 to 5 miles.
C or D **C** Domain **[0, 5]** Why? **any distance is possible**

2. A vending machine sells 0 to 6 drinks.
C or D **D** Domain **{ 0, 1, 2, 3, 4, 5, 6 }** Why? **whole drinks only**

3. Water in a bathtub drops from 30 gallons to 0.
C or D **C** Range **[0, 30]** Why? **volume changes smoothly**

4. A class of 24 students is split into equal teams of 4.
C or D **D** Domain (teams) **{ 1, 2, 3, 4, 5, 6 }**

5. Explain in your own words why a discrete domain cannot use interval notation.

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Exit Ticket

A school orders pizzas for a party. Each pizza feeds 6 students, and they order up to 5 pizzas.

Domain (pizzas) **{ 0, 1, 2, 3, 4, 5 }** Range (students fed) **{ 0, 6, 12, 18, 24, 30 }**

Continuous or discrete, and why? **discrete - whole pizzas only**